Key

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f26.207.quiz09::30

In all quizzes and exams, show your work and write with correct notation.

(10pts) 1. Where is $f(x) = x^3 - 3x$ increasing/decreasing?

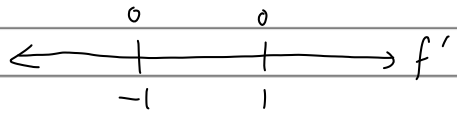
$f'(x) = 3x^2 - 3$ always defined

$3x^2 - 3 = 0$

$3(x^2 - 1) = 0$

$3(x+1)(x-1) = 0$

$3 = 0$ no sol. $x+1=0$, $x = -1$ $x-1=0$, $x = 1$



$f'(x) = 3(x+1)(x-1)$

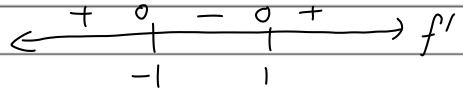
Why? f' gives slope. We need slope info to tell if f is increasing or decreasing.

Must evaluate $f'(x)$ not $f(x)$

$f'(-2)$ is $(+)(-)(-)$ is $+$

$f'(0)$ is $(+)(+)(-)$ is $-$

$f'(2)$ is $(+)(+)(+)$ is $+$



f is increasing on $(-\infty, -1) \cup (1, \infty)$

f is decreasing on $(-1, 1)$

(10pts) 2. If $f(2) = 20$ and $f'(x) \leq 2$ for all x, how big can $f(6)$ possibly be?

State by name

f is differentiable for all x, so f is continuous for all x.

Since f is continuous on $[2,6]$ and differentiable on $(2,6)$, by the Mean Value Theorem there exists an x-value c in $(2,6)$ such that $f'(c) = \frac{f(6)-f(2)}{6-2}$

$$\frac{f(6)-f(2)}{6-2} = f'(c) \leq 2$$

$$\frac{f(6)-20}{4} \leq 2$$

$$f(6) - 20 \leq 8$$

$$f(6) \leq 28$$

(10pts) 3. Find the linearization of $f(x) = x^2$ at $x = 3$

$f'(x) = 2x$

$f'(3) = 2(3) = 6$

$f(3) = 3^2 = 9$

Need 6 here, not $2x$

$y - 9 = 6(x-3)$

$y = 6(x-3) + 9$

$L(x) = 6(x-3) + 9$